NET732

Week 5:Network Intrusion Analysis

Practical

1. Installing Nmap

I opened Ubuntu Terminal and installed Nmap using:

sudo apt update

sudo apt install nmap -y

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week5$ nmap  --version
Nmap version 7.98 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.5.5 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week5$
```

I checked that the installation was successful using:

nmap –version

2. Checking my IP address

I used the following command to check my Ubuntu IP address:

ip addr

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week5$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 00:0c:29:4a:af:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.134.128/24 brd 192.168.134.255 scope global dynamic noprefixroute ens33
       valid_lft 1611sec preferred_lft 1611sec
    inet6 fe80::20c:29ff:fe4a:af9c/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week5$
```

This helped me identify my active network interface and IP address.

3. Performing a network scan

I used Nmap to scan my Ubuntu machine:

nmap -sV 127.0.0.1

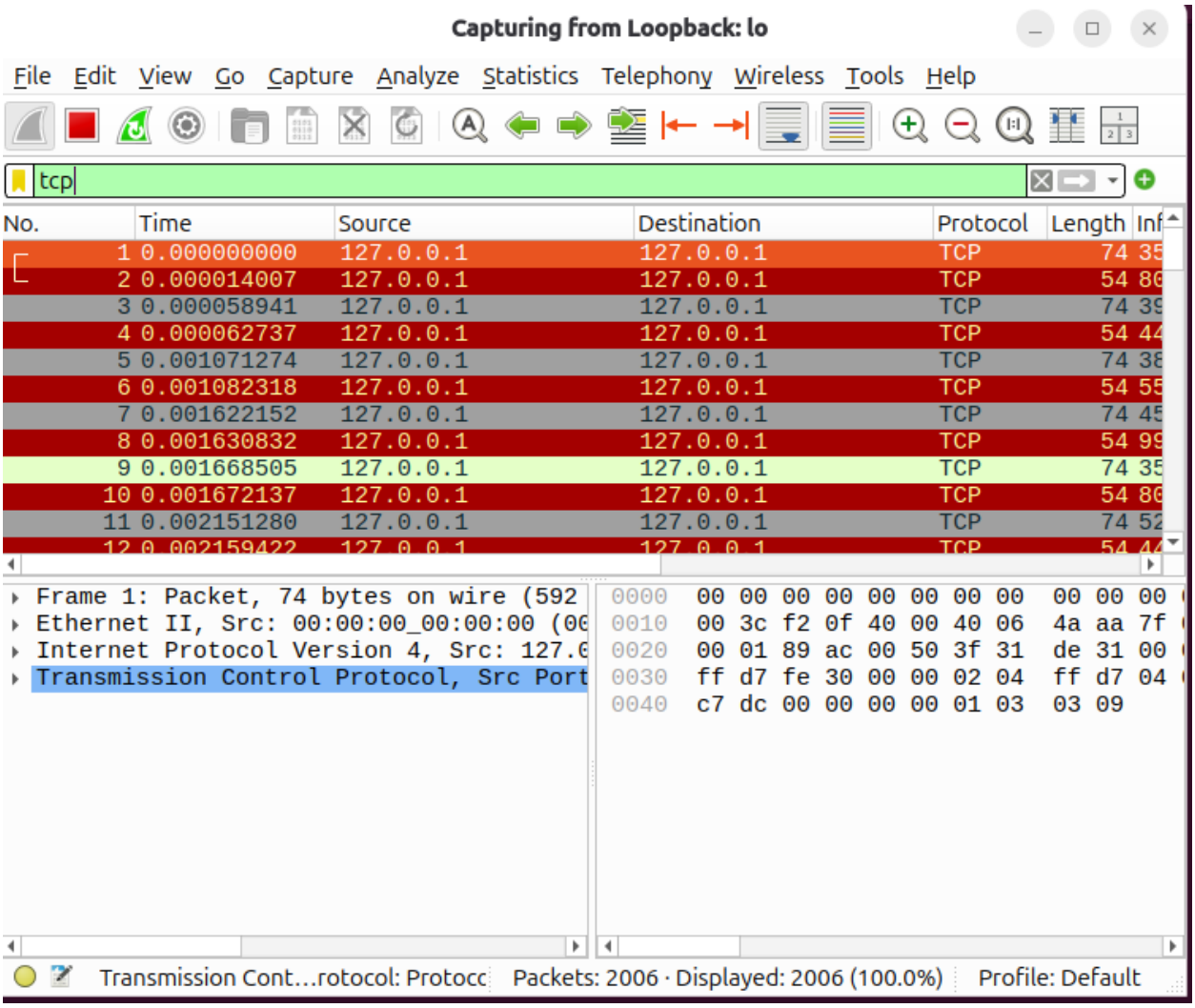
```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week5$ nmap -sV 127 .0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-30 17:35 +0200
Failed to resolve "127".
Failed to resolve ".0.0.1".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 7.29 seconds
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week5$
```

The scan helped me identify open ports and services running on my machine.

4. Capturing network traffic

I opened Wireshark and selected the Loopback interface. I started capturing packets and ran the following command:

nmap -sT -p 1-1000 127.0.0.1




I used the TCP filter in Wireshark to view the packets generated by the scan.

5. Saving scan results

I saved my Nmap scan results using:

```
nmap -sV 127.0.0.1 | tee nmap_results.txt
```

6. Analysis and conclusion

During this practical, I used Nmap to scan my Ubuntu machine and identify open ports and services. I also used Wireshark to capture and analyse the TCP traffic generated during the scan.

I learned that Nmap is useful for identifying network services, while Wireshark helps analyse network communication.

These tools are important in SOC operations because security analysts use them to investigate scanning activities and identify possible security threats.

I successfully completed the network scanning practical and saved my evidence. My next step is to practise identifying suspicious scanning patterns and investigating unusual network traffic.